# Entanglement Entropy in Conformal Field Theory

Replica Trick, Twist Fields, and the Formula $S = \frac{c}{3} \ln L$

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## Contents

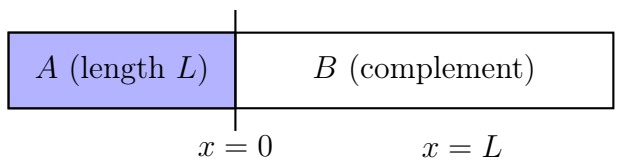
# 1 Introduction

Entanglement entropy is a fundamental measure of quantum correlations in many-body systems. In conformal field theory (CFT), it takes a universal logarithmic form that depends only on the central charge $c$, making it a powerful tool for characterizing quantum phases and critical points.

## 1.1 Physical Setup

Consider a 1+1D quantum system in its ground state $|0\rangle$. We partition the system into two regions:

- Region $A$: interval of length $L$ (the subsystem we study)
- Region $B$: the complement (traced out)




## 1.2 Entanglement Entropy Definition

The reduced density matrix is obtained by tracing out region $B$:

$$\rho_A = \mathrm{Tr}_B \, |0\rangle \langle 0| \tag{1}$$

The von Neumann entanglement entropy is:

$$S_A = -\,\mathrm{Tr}(\rho_A \ln \rho_A) \tag{2}$$

## 1.3 Physical Interpretation

| **Value** | **Interpretation** | **Example** |
|---|---|---|
| $S_A = 0$ | Product state (no entanglement) | $\|\psi\rangle_A \otimes \|\phi\rangle_B$ |
| $S_A > 0$ | Entangled state | Bell state, GHZ state |
| $S_A \propto L$ | Volume law (thermal states) | High-energy eigenstates |
| $S_A \propto \ln L$ | Area law in 1D (CFT ground states) | Critical systems |

## 1.4 Main Result

For a 1+1D CFT with central charge $c$, the entanglement entropy of an interval of length $L$ is:

$$S_A = \frac{c}{3} \ln \left( \frac{L}{a} \right) + \text{const} \tag{3}$$

where $a$ is the UV cutoff (lattice spacing).

**For the Ising model ($c = 1/2$):**

$$S_A = \frac{1}{6} \ln \left( \frac{L}{a} \right) + \text{const} \tag{4}$$

# 2 The Replica Trick: Avoiding Direct Diagonalization

## 2.1 The Fundamental Challenge

The entanglement entropy is defined as:

$$S_A = -\operatorname{Tr}(\rho_A \ln \rho_A) \tag{5}$$

To compute this directly, we face a formidable obstacle:

- **Step 1:** Diagonalize the reduced density matrix $\rho_A$ to find all eigenvalues $\{\lambda_i\}$
- **Step 2:** Compute the sum: $S_A = -\sum_i \lambda_i \ln \lambda_i$

**Why is this hard?**
For a quantum field theory or many-body system:

- The Hilbert space dimension grows *exponentially* with subsystem size: $\dim(\mathcal{H}_A) \sim d^L$
- $\rho_A$ is typically a dense, high-dimensional matrix
- Finding eigenvalues of an exponential-sized matrix is computationally intractable
- Even if we could diagonalize $\rho_A$, the eigenvalues themselves are complicated functionals of the field configuration

**Example:** For a chain of $L = 100$ spin-1/2 particles:

$$\dim(\mathcal{H}_A) = 2^{100} \approx 10^{30} \tag{6}$$

Diagonalizing a $10^{30} \times 10^{30}$ matrix is impossible!

## 2.2 Rényi Entropies

Instead of attacking $S_A$ directly, we compute a family of related quantities.

**Definition:** The **Rényi entropies** are:

$$S_A^{(n)} = \frac{1}{1-n} \ln \operatorname{Tr}(\rho_A^n) \tag{7}$$

where $n$ is a positive integer ($n = 2, 3, 4, \ldots$).

In path integral language, $\operatorname{Tr}(\rho_A^n)$ has a geometric interpretation:

- $\operatorname{Tr}(\rho_A^n)$ involves *multiplying* density matrices, not diagonalizing them
- We can compute $\operatorname{Tr}(\rho_A^n)$ for integer $n$, then analytically continue to real $n$
- The limit $n \to 1$ recovers the von Neumann entropy

$\mathrm{Tr}(\rho_A^n)$ is called the *purity* (for $n = 2$) or *Rényi purity* (for general $n$).

**Proof:** Let $\{\lambda_i\}$ be the eigenvalues of $\rho_A$. Since $\rho_A$ is a density matrix:

$$\sum_i \lambda_i = 1, \quad 0 \leq \lambda_i \leq 1 \tag{8}$$

For $n = 2$:

$$\mathrm{Tr}\left(\rho_A^2\right) = \sum_i \lambda_i^2 \tag{9}$$

**Pure state:** A pure state satisfies $\rho_A = |\psi\rangle\langle\psi|$, which is a rank-1 projector:

$$\rho_A^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi| = \rho_A \tag{10}$$

Taking the trace:

$$\mathrm{Tr}\left(\rho_A^2\right) = \mathrm{Tr}(\rho_A) = 1 \tag{11}$$

In the eigenbasis, this corresponds to one eigenvalue $\lambda_1 = 1$ and all others $\lambda_{i>1} = 0$.

**Mixed state:** A mixed state is a convex combination of pure states:

$$\rho_A = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad \sum_i p_i = 1, \quad 0 < p_i < 1 \tag{12}$$

The eigenvalues are $\lambda_i = p_i$. Since $0 < \lambda_i < 1$ for at least two eigenvalues:

$$\lambda_i^2 < \lambda_i \quad \text{(strict inequality for each } 0 < \lambda_i < 1) \tag{13}$$

Summing over all eigenvalues:

$$\mathrm{Tr}\left(\rho_A^2\right) = \sum_i \lambda_i^2 < \sum_i \lambda_i = 1 \tag{14}$$

The inequality is strict because at least two eigenvalues contribute.

**Entanglement measure:** For a system entangled with environment, $\rho_A$ is mixed. More entanglement $\Rightarrow$ more uniform eigenvalue distribution $\Rightarrow$ smaller $\mathrm{Tr}(\rho_A^n)$:

- $\mathrm{Tr}(\rho_A^2) = 1$: pure state (no entanglement)
- $\mathrm{Tr}(\rho_A^2) < 1$: mixed state (entangled with environment)
- Smaller $\mathrm{Tr}(\rho_A^n) \Rightarrow$ more entanglement

## 2.3 Recovering von Neumann Entropy

The key claim is:

$$S_A = \lim_{n \to 1} S_A^{(n)} \tag{15}$$

Let $\{\lambda_i\}$ be the eigenvalues of $\rho_A$. Then:

$$\mathrm{Tr}(\rho_A^n) = \sum_i \lambda_i^n \tag{16}$$

Define $f(n) = \ln \mathrm{Tr}(\rho_A^n) = \ln\left(\sum_i \lambda_i^n\right)$.
The Rényi entropy is:

$$S_A^{(n)} = \frac{f(n)}{1-n} \tag{17}$$

Taking $n \to 1$, we get 0/0 indeterminate form. Apply L'Hôpital's rule:

$$S_A = \lim_{n\to 1} \frac{f(n)}{1-n} = -\left.\frac{df}{dn}\right|_{n=1} \tag{18}$$

Compute the derivative:

$$\frac{df}{dn} = \frac{d}{dn} \ln\left(\sum_i \lambda_i^n\right) \tag{19}$$

$$= \frac{\sum_i \lambda_i^n \ln \lambda_i}{\sum_i \lambda_i^n} \tag{20}$$

At $n = 1$:

$$\left.\frac{df}{dn}\right|_{n=1} = \frac{\sum_i \lambda_i \ln \lambda_i}{\sum_i \lambda_i} = \sum_i \lambda_i \ln \lambda_i = -S_A \tag{21}$$

Therefore:

$$S_A = -(-S_A) = S_A \tag{22}$$

## 2.4 Physical Picture

To compute $\mathrm{Tr}(\rho_A^n)$:

1. Take $n$ *copies* (replicas) of the system
2. Sew them together in a cyclic pattern
3. Compute a partition function on this $n$-sheeted geometry
4. Take $n \to 1$ at the end

The name "replica" comes from step 1.
**Historical note:** This method was borrowed from spin glass theory.

## 2.5 Workflow

1. **Compute** $\mathrm{Tr}(\rho_A^n)$ for integer $n \geq 2$
   - Use path integral representation
   - Geometric interpretation: $n$-sheeted Riemann surface
2. **Analytically continue** to real $n$
   - Treat $n$ as a continuous parameter
   - Assume smooth behavior near $n = 1$
3. **Take the limit** $n \to 1$
   - Apply L'Hôpital's rule
   - Recover von Neumann entropy

# 3 Path Integral Representation

## 3.1 Trace Formula in Field Basis

The trace of $\rho_A^n$ can be expressed as a functional integral over field configurations in region $A$.

**Statement:**

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A \, \langle \phi_A | \rho_A^n | \phi_A \rangle \tag{23}$$

where:

- Tr = Trace over Hilbert space $\mathcal{H}_A$
- $\int \mathcal{D}\phi_A$ = Functional integral over all field configurations in region $A$
- $|\phi_A\rangle$ = Field eigenstate in region $A$

**Proof:**

**Step 1:** Definition of trace. In any orthonormal basis $\{|k\rangle\}$ of $\mathcal{H}_A$:

$$\mathrm{Tr}(\rho_A^n) = \sum_k \langle k | \rho_A^n | k \rangle \tag{24}$$

**Step 2:** Choose field eigenstate basis. Field eigenstates in region $A$:

$$\hat{\phi}(x)|\phi_A\rangle = \phi_A(x)|\phi_A\rangle \quad \text{for } x \in A \tag{25}$$

**Step 3:** Completeness relation. For field eigenstates:

$$\int \mathcal{D}\phi_A \, |\phi_A\rangle\langle\phi_A| = I_A \quad \text{(identity on } \mathcal{H}_A\text{)} \tag{26}$$

The measure $\mathcal{D}\phi_A$ is defined by discretizing space:

$$A = \{x_1, x_2, \ldots, x_M\}, \quad \phi_A = \{\phi_1, \phi_2, \ldots, \phi_M\} \tag{27}$$

$$\mathcal{D}\phi_A = \prod_{j=1}^{M} d\phi_j \quad \xrightarrow{M\to\infty} \quad \text{functional measure} \tag{28}$$

**Step 4:** Orthogonality. Field eigenstates are orthogonal:

$$\langle \phi_A | \phi_A' \rangle = \delta[\phi_A - \phi_A'] = \prod_{x\in A} \delta(\phi_A(x) - \phi_A'(x)) \tag{29}$$

**Step 5:** Insert completeness into trace:

$$\mathrm{Tr}(\rho_A^n) = \sum_k \langle k | \rho_A^n \left( \int \mathcal{D}\phi_A \, |\phi_A\rangle\langle\phi_A| \right) |k\rangle \tag{30}$$

$$= \int \mathcal{D}\phi_A \sum_k \langle k | \rho_A^n | \phi_A \rangle \langle \phi_A | k \rangle \tag{31}$$

**Step 6:** Swap sum and integral. Using $\sum_k |k\rangle\langle k| = I_A$:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A \sum_k \langle\phi_A|k\rangle\langle k|\rho_A^n|\phi_A\rangle \tag{32}$$

$$= \int \mathcal{D}\phi_A \, \langle\phi_A| \left(\sum_k |k\rangle\langle k|\right) \rho_A^n|\phi_A\rangle \tag{33}$$

$$= \int \mathcal{D}\phi_A \, \langle\phi_A|\rho_A^n|\phi_A\rangle \tag{34}$$

**Step 7:** Final result:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A \, \langle\phi_A|\rho_A^n|\phi_A\rangle \tag{35}$$

## 3.2 Alternative Derivation: Direct from Matrix Elements

In the field representation, $\rho_A^n$ has matrix elements:

$$(\rho_A^n)(\phi_A, \phi'_A) = \langle\phi_A|\rho_A^n|\phi'_A\rangle \tag{36}$$

For a discrete basis, the trace is the sum of diagonal elements:

$$\mathrm{Tr}\left(\hat{O}\right) = \sum_i O_{ii} \tag{37}$$

For a continuous basis (field configurations), the sum becomes a functional integral:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A \, \langle\phi_A|\rho_A^n|\phi_A\rangle \tag{38}$$

## 3.3 Lattice Regularization

To make the functional integral well-defined, we discretize space.

**Lattice setup:**

- Region $A$: $M$ lattice sites $\{x_1, x_2, \ldots, x_M\}$
- Field configuration: $\vec{\phi} = (\phi_1, \phi_2, \ldots, \phi_M)$ where $\phi_j = \phi(x_j)$
- Hilbert space: $\mathcal{H}_A = L^2(\mathbb{R}^M)$ (square-integrable functions of $\vec{\phi}$)

**Field eigenstates on lattice:**

$$|\vec{\phi}\rangle \quad \text{where} \quad \hat{\phi}_j|\vec{\phi}\rangle = \phi_j|\vec{\phi}\rangle \tag{39}$$

**Completeness:**

$$\int d^M\phi \, |\vec{\phi}\rangle\langle\vec{\phi}| = I_A, \quad d^M\phi = d\phi_1 \, d\phi_2 \cdots d\phi_M \tag{40}$$

**Orthogonality:**

$$\langle\vec{\phi}|\vec{\phi}'\rangle = \delta^M(\vec{\phi}-\vec{\phi}') = \prod_{j=1}^{M}\delta(\phi_j-\phi_j') \tag{41}$$

**Trace on lattice:**

$$\begin{aligned}
\mathrm{Tr}(\rho_A^n) &= \sum_k \langle k|\rho_A^n|k\rangle && (42)\\
&= \sum_k \langle k|\rho_A^n \left(\int d^M\phi\, |\vec{\phi}\rangle\langle\vec{\phi}|\right)|k\rangle && (43)\\
&= \int d^M\phi \sum_k \langle\vec{\phi}|k\rangle\langle k|\rho_A^n|\vec{\phi}\rangle && (44)\\
&= \int d^M\phi\, \langle\vec{\phi}|\rho_A^n|\vec{\phi}\rangle && (45)
\end{aligned}$$

**Continuum limit ($M\to\infty$):**

$$\int d^M\phi \to \int \mathcal{D}\phi_A, \quad |\vec{\phi}\rangle \to |\phi_A\rangle \tag{46}$$

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi_A\, \langle\phi_A|\rho_A^n|\phi_A\rangle \tag{47}$$

## 3.4 Euclidean Path Integral for Ground State

**Ground state preparation:**

$$|0\rangle = \lim_{T\to\infty} e^{-HT}|\text{arbitrary}\rangle \tag{48}$$

In the path integral formalism, this corresponds to evolving from $\tau=-\infty$ to $\tau=0$.

**Derivation:**

Consider the transition amplitude from an arbitrary state $|\psi\rangle$ at $\tau=-T$ to a field configuration $|\phi_f\rangle$ at $\tau=0$:

$$\langle\phi_f|e^{-HT}|\psi\rangle = \int_{\phi(-T)=\psi}^{\phi(0)=\phi_f} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{49}$$

Expand $|\psi\rangle$ in the energy eigenbasis:

$$|\psi\rangle = c_0|0\rangle + \sum_{n>0} c_n|n\rangle \tag{50}$$

Apply Euclidean time evolution:

$$\begin{aligned}
e^{-HT}|\psi\rangle &= c_0 e^{-E_0T}|0\rangle + \sum_{n>0} c_n e^{-E_nT}|n\rangle && (51)\\
&= c_0 e^{-E_0T}\left[|0\rangle + \sum_{n>0}\frac{c_n}{c_0}e^{-(E_n-E_0)T}|n\rangle\right] && (52)
\end{aligned}$$

Since $E_n > E_0$ for all excited states, as $T \to \infty$:

$$e^{-(E_n - E_0)T} \to 0 \quad \text{for all } n > 0 \tag{53}$$

Therefore:

$$\lim_{T\to\infty} e^{-HT}|\psi\rangle = c_0 e^{-E_0 T}|0\rangle \tag{54}$$

The ground state is projected out. Normalizing:

$$|0\rangle = \lim_{T\to\infty} \frac{e^{-HT}|\psi\rangle}{\sqrt{\langle\psi|e^{-2HT}|\psi\rangle}} \tag{55}$$

**Path integral representation:**

$$\langle\phi_f|0\rangle = \frac{1}{\sqrt{Z}} \int_{\phi(-\infty)=\text{arbitrary}}^{\phi(0)=\phi_f} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{56}$$

The ground state wavefunctional is obtained by integrating over all field configurations from $\tau = -\infty$ to $\tau = 0$ with boundary condition $\phi(0) = \phi_f$.

### 3.4.1 Definition of the Functional Measure

**What is $\mathcal{D}\phi$?**

The functional measure $\mathcal{D}\phi$ (also called the path integral measure) is the continuum limit of a product of ordinary integrals over field values at each spacetime point.

**Lattice regularization:**

Discretize spacetime into a lattice with:

- Spatial sites: $x_j = ja$, $j = 1, \ldots, N_s$ (lattice spacing $a$)
- Time slices: $\tau_k = k\epsilon$, $k = 1, \ldots, N_\tau$ (time step $\epsilon$)

The field becomes a set of discrete variables:

$$\phi(x,\tau) \quad \to \quad \phi_{j,k} \equiv \phi(x_j, \tau_k) \tag{57}$$

**Functional measure as a product:**

$$\mathcal{D}\phi \quad \equiv \quad \lim_{a\to 0,\, \epsilon\to 0} \prod_{j=1}^{N_s} \prod_{k=1}^{N_\tau} d\phi_{j,k} \tag{58}$$

Each $d\phi_{j,k}$ is an ordinary Lebesgue integral over the real line (assuming a real scalar field).

**Normalization:**

The measure includes a normalization factor to ensure convergence:

$$\mathcal{D}\phi = \mathcal{N} \prod_{j,k} d\phi_{j,k} \tag{59}$$

where $\mathcal{N}$ is chosen such that:

$$\int \mathcal{D}\phi\, e^{-S_E[\phi]} = Z \tag{60}$$

For a free scalar field, $\mathcal{N}$ is determined by Gaussian integration.

**Why is this well-defined?**

The functional integral is defined as the limit of well-defined finite-dimensional integrals:

$$\int \mathcal{D}\phi\, F[\phi] = \lim_{N_s, N_\tau \to \infty} \int \left( \prod_{j,k} d\phi_{j,k} \right) F(\{\phi_{j,k}\}) \tag{61}$$

For free (Gaussian) theories, the limit exists and can be computed exactly. For interacting theories, the measure is defined perturbatively or via lattice regularization.

**Analogy with quantum mechanics:**

For a single particle:

$$\int \mathcal{D}q(\tau) = \lim_{N \to \infty} \int dq_1 \cdots dq_{N-1} \tag{62}$$

where $q_k = q(\tau_k)$ and the endpoints $q_0$, $q_N$ are fixed by boundary conditions.

**In the replica trick:**

The measure $\mathcal{D}\phi_A$ for the reduced density matrix trace is:

$$\int \mathcal{D}\phi_A = \int \prod_{x \in A} d\phi_A(x) \tag{63}$$

integrating over all field configurations in region $A$ only.

## 3.5 Path Integral Derivation: Quantum Mechanics

**System:** Single particle in 1D with Hamiltonian:

$$H = \frac{\hat{p}^2}{2m} + V(\hat{q}) \tag{64}$$

**Goal:** Compute transition amplitude in Euclidean time:

$$\langle q_f | e^{-\beta H} | q_i \rangle \tag{65}$$

where $\beta = 1/T$ (inverse temperature).

### 3.5.1 Trotter Decomposition

Divide time into $N$ slices:

$$\epsilon = \beta/N, \quad e^{-\beta H} = (e^{-\epsilon H})^N \tag{66}$$

Insert $N-1$ complete sets of position states:

$$\langle q_f | e^{-\beta H} | q_i \rangle = \int dq_1 \cdots dq_{N-1} \prod_{k=0}^{N-1} \langle q_{k+1} | e^{-\epsilon H} | q_k \rangle \tag{67}$$

where $q_0 = q_i$, $q_N = q_f$.

### 3.5.2 Short-Time Propagator

For small $\epsilon$, use Trotter formula:

$$e^{-\epsilon H} = e^{-\epsilon(\hat{p}^2/(2m)+V(\hat{q}))} \approx e^{-\epsilon\hat{p}^2/(2m)} e^{-\epsilon V(\hat{q})} + O(\epsilon^2) \tag{68}$$

Insert momentum eigenstates:

$$\langle q_{k+1}|e^{-\epsilon H}|q_k\rangle \approx \int \frac{dp_k}{2\pi} \langle q_{k+1}|e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle\langle p_k|e^{-\epsilon V(\hat{q})}|q_k\rangle \tag{69}$$

**Origin of the $1/(2\pi)$ factor:**

The completeness relation for momentum eigenstates depends on the normalization convention. With $\langle p|q\rangle = e^{-ipq}$ (no prefactor):

$$\langle p|p'\rangle = 2\pi\delta(p-p') \tag{70}$$

The completeness relation is:

$$\int \frac{dp}{2\pi} |p\rangle\langle p| = I \tag{71}$$

The $1/(2\pi)$ ensures proper normalization. Equivalently, one can use $\langle p|q\rangle = \frac{1}{\sqrt{2\pi}} e^{-ipq}$ with $\int dp\, |p\rangle\langle p| = I$.

**Derivation of matrix elements:**

*First factor:* $\langle p_k|e^{-\epsilon V(\hat{q})}|q_k\rangle$

**Lemma:** If $V(\hat{q})$ is a function of the position operator, it acts diagonally on position eigenstates:

$$V(\hat{q})|q\rangle = V(q)|q\rangle \tag{72}$$

*Proof:*

**Step 1: Definition of function of an operator**

A function $V(\hat{q})$ of an operator is defined via its power series expansion (assuming $V$ is analytic):

$$V(\hat{q}) = \sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!} \hat{q}^n \tag{73}$$

where $V^{(n)}(0)$ is the $n$-th derivative of $V$ evaluated at 0.

**Step 2: Action of powers of $\hat{q}$ on eigenstates**

By definition of the position eigenstate:

$$\hat{q}|q\rangle = q|q\rangle \tag{74}$$

Apply $\hat{q}$ again:

$$\hat{q}^2|q\rangle = \hat{q}(\hat{q}|q\rangle) = \hat{q}(q|q\rangle) = q(\hat{q}|q\rangle) = q^2|q\rangle \tag{75}$$

By induction, for any integer $n \geq 0$:

$$\hat{q}^n|q\rangle = q^n|q\rangle \tag{76}$$

**Step 3: Action of $V(\hat{q})$ on eigenstates**

Apply the power series to $|q\rangle$:

$$V(\hat{q})|q\rangle = \left(\sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!}\hat{q}^n\right)|q\rangle \tag{77}$$

$$= \sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!}\hat{q}^n|q\rangle \tag{78}$$

$$= \sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!}q^n|q\rangle \tag{79}$$

$$= \left(\sum_{n=0}^{\infty} \frac{V^{(n)}(0)}{n!}q^n\right)|q\rangle \tag{80}$$

$$= V(q)|q\rangle \tag{81}$$

where the last line recognizes the power series for $V(q)$.

**Alternative proof (spectral theorem):**

For any Hermitian operator $\hat{q}$ with continuous spectrum, the spectral theorem gives:

$$\hat{q} = \int dq\, q\, |q\rangle\langle q| \tag{82}$$

A function of the operator is:

$$V(\hat{q}) = \int dq\, V(q)\, |q\rangle\langle q| \tag{83}$$

Apply to a position eigenstate $|q_0\rangle$:

$$V(\hat{q})|q_0\rangle = \int dq\, V(q)\, |q\rangle\langle q|q_0\rangle \tag{84}$$

$$= \int dq\, V(q)\, |q\rangle\delta(q - q_0) \tag{85}$$

$$= V(q_0)|q_0\rangle \tag{86}$$

This proves the lemma. □

**Step 4: Exponential of $V(\hat{q})$**

Since $e^{-\epsilon V(\hat{q})}$ is a function of $\hat{q}$ (with $f(x) = e^{-\epsilon V(x)}$), the lemma applies:

$$e^{-\epsilon V(\hat{q})}|q_k\rangle = e^{-\epsilon V(q_k)}|q_k\rangle \tag{87}$$

**Step 5: Matrix element**

Now take the inner product with $\langle p_k|$:

$$\langle p_k|e^{-\epsilon V(\hat{q})}|q_k\rangle = e^{-\epsilon V(q_k)}\langle p_k|q_k\rangle \tag{88}$$

Using the position-momentum overlap $\langle p|q\rangle = e^{-ipq}$ (derived in Appendix A):

$$\boxed{\langle p_k|e^{-\epsilon V(\hat{q})}|q_k\rangle = e^{-\epsilon V(q_k)}e^{-ip_k q_k}} \tag{89}$$

*Second factor:* $\langle q_{k+1}|e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle$
Since $\hat{p}^2$ acts diagonally on momentum eigenstates:

$$\hat{p}^2|p_k\rangle = p_k^2|p_k\rangle \tag{90}$$

Therefore:

$$e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle = e^{-\epsilon p_k^2/(2m)}|p_k\rangle \tag{91}$$

Now take the inner product with $\langle q_{k+1}|$:

$$\langle q_{k+1}|e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle = e^{-\epsilon p_k^2/(2m)}\langle q_{k+1}|p_k\rangle \tag{92}$$

Using $\langle q|p\rangle = e^{ipq}$:

$$\langle q_{k+1}|e^{-\epsilon\hat{p}^2/(2m)}|p_k\rangle = e^{-\epsilon p_k^2/(2m)}e^{ip_k q_{k+1}} \tag{93}$$

Combine:

$$\langle q_{k+1}|e^{-\epsilon H}|q_k\rangle = \int \frac{dp_k}{2\pi}\exp\left[-\frac{\epsilon p_k^2}{2m} + ip_k(q_{k+1}-q_k) - \epsilon V(q_k)\right] \tag{94}$$

### 3.5.3 Gaussian Momentum Integral

Complete the square:

$$\int \frac{dp}{2\pi}\exp\left[-\frac{\epsilon p^2}{2m} + ip\Delta q\right] = \sqrt{\frac{m}{2\pi\epsilon}}\exp\left[-\frac{m(\Delta q)^2}{2\epsilon}\right] \tag{95}$$

Result:

$$\langle q_{k+1}|e^{-\epsilon H}|q_k\rangle = \sqrt{\frac{m}{2\pi\epsilon}}\exp\left[-\epsilon\frac{m}{2}\left(\frac{q_{k+1}-q_k}{\epsilon}\right)^2 - \epsilon V(q_k)\right] \tag{96}$$

### 3.5.4 Continuum Limit

Take $N \to \infty$, $\epsilon \to 0$:

$$\frac{q_{k+1}-q_k}{\epsilon} \to \frac{dq}{d\tau} \equiv \dot{q}(\tau) \tag{97}$$

$$\epsilon\sum_k \to \int_0^\beta d\tau \tag{98}$$

$$\prod_k dq_k\sqrt{\frac{m}{2\pi\epsilon}} \to \mathcal{D}q \text{ (functional measure)} \tag{99}$$

Final path integral:

$$\langle q_f|e^{-\beta H}|q_i\rangle = \int_{q(0)=q_i}^{q(\beta)=q_f}\mathcal{D}q\, e^{-S_E[q]} \tag{100}$$

where Euclidean action:

$$S_E[q] = \int_0^\beta d\tau\left[\frac{1}{2}m\dot{q}^2 + V(q)\right] \tag{101}$$

## 3.6 Path Integral Derivation: Quantum Field Theory

### 3.6.1 Lattice Discretization

Space: Discretize into $M$ sites with spacing $a$:

$$x_j = j \cdot a \quad \text{for } j = 0, 1, \ldots, M-1 \tag{102}$$

Field at site $j$: $\phi_j(\tau) \equiv \phi(x_j, \tau)$

Conjugate momentum: $\pi_j(\tau)$

Hamiltonian:

$$H = \sum_{j=0}^{M-1} a \left[ \frac{\pi_j^2}{2} + \frac{(\partial_x \phi_j)^2}{2} + V(\phi_j) \right] \tag{103}$$

where $(\partial_x \phi_j) \approx (\phi_{j+1} - \phi_j)/a$.

### 3.6.2 Field Eigenstates

Define:

$$|\phi\rangle \equiv |\{\phi_j\}\rangle = |\phi_0\rangle \otimes |\phi_1\rangle \otimes \cdots \otimes |\phi_{M-1}\rangle \tag{104}$$

Eigenvalue equation:

$$\hat{\phi}_j |\phi_k\rangle = \phi_j \delta_{jk} |\phi_k\rangle \tag{105}$$

Orthogonality:

$$\langle \phi | \phi' \rangle = \prod_{j=0}^{M-1} \delta(\phi_j - \phi'_j) \tag{106}$$

Completeness:

$$\int \mathcal{D}\phi \, |\phi\rangle\langle\phi| = I, \quad \mathcal{D}\phi = \prod_{j=0}^{M-1} d\phi_j \tag{107}$$

### 3.6.3 Continuum Limit

Take $a \to 0$:

$$\sum_j a \to \int dx \tag{108}$$

$$\phi_j(\tau) \to \phi(x, \tau) \tag{109}$$

$$\pi_j(\tau) \to \pi(x, \tau) \tag{110}$$

Euclidean action:

$$S_E[\phi] = \int_0^\beta d\tau \int dx \left[ \frac{1}{2}(\partial_\tau \phi)^2 + \frac{1}{2}(\partial_x \phi)^2 + V(\phi) \right] \tag{111}$$

Path integral:

$$\langle \phi_f | e^{-\beta H} | \phi_i \rangle = \int_{\phi(x,0)=\phi_i(x)}^{\phi(x,\beta)=\phi_f(x)} \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{112}$$

## 3.7 Thermal Density Matrix

**Thermal density operator:**

$$\hat{\rho} = \frac{e^{-\beta H}}{Z}, \quad Z = \mathrm{Tr}\big(e^{-\beta H}\big) \tag{113}$$

**Matrix elements in field basis:**

$$\rho(\phi_f|\phi_i) = \langle\phi_f|\hat{\rho}|\phi_i\rangle = \frac{1}{Z}\langle\phi_f|e^{-\beta H}|\phi_i\rangle \tag{114}$$

Path integral representation:

$$\rho(\phi_f|\phi_i) = \frac{1}{Z}\int_{\phi(x,0)=\phi_i(x)}^{\phi(x,\beta)=\phi_f(x)} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{115}$$

**Delta function form (Eq. 7 of Calabrese-Cardy):**

$$\rho(\phi_f|\phi_i) = \frac{1}{Z}\int \mathcal{D}\phi\, e^{-S_E[\phi]} \prod_x \delta(\phi(x,0)-\phi_i(x))\delta(\phi(x,\beta)-\phi_f(x)) \tag{116}$$

**Interpretation:** The delta function product enforces boundary conditions on the path integral:

- $\prod_x \delta(\phi(x,0)-\phi_i(x))$: fixes the *initial condition* at $\tau = 0$ to be $\phi_i(x)$
- $\prod_x \delta(\phi(x,\beta)-\phi_f(x))$: fixes the *final condition* at $\tau = \beta$ to be $\phi_f(x)$
- The functional integral $\int \mathcal{D}\phi$ is now over *all* field configurations without boundary restrictions
- The delta functions select only those configurations satisfying the boundary conditions

**Equivalence:** This form is equivalent to the boundary-condition notation:

$$\int_{\phi(x,0)=\phi_i(x)}^{\phi(x,\beta)=\phi_f(x)} \mathcal{D}\phi\, e^{-S_E[\phi]} = \int \mathcal{D}\phi\, e^{-S_E[\phi]} \prod_x \delta(\phi(x,0)-\phi_i(x))\delta(\phi(x,\beta)-\phi_f(x)) \tag{117}$$

The delta function form is useful when computing traces or partial traces, as it makes the boundary conditions explicit in the integrand.

**Partition function as trace:**

$$Z = \mathrm{Tr}\big(e^{-\beta H}\big) = \int \mathcal{D}\phi\, \langle\phi|e^{-\beta H}|\phi\rangle = \int_{\phi(x,0)=\phi(x,\beta)} \mathcal{D}\phi\, e^{-S_E[\phi]} \tag{118}$$

Geometry: Cylinder with circumference $\beta$.

## 3.8 Reduced Density Matrix and Partial Trace

**Spatial partition:**

$$\text{Space} = A \cup B \quad \text{(disjoint)} \tag{119}$$

where $A$ is the subsystem of interest (e.g., interval $[u, v]$) and $B$ is the complement.

Hilbert space factorization:

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B, \quad |\phi\rangle = |\phi_A\rangle \otimes |\phi_B\rangle \tag{120}$$

**Reduced density matrix:**

$$\rho_A = \mathrm{Tr}_B(\rho) \tag{121}$$

Matrix elements:

$$\langle \phi_A | \rho_A | \phi'_A \rangle = \int \mathcal{D}\phi_B \, \langle \phi_A, \phi_B | \rho | \phi'_A, \phi_B \rangle \tag{122}$$

**Path integral derivation:**

$$\langle \phi_A | \rho_A | \phi'_A \rangle = \frac{1}{Z} \int \mathcal{D}\phi_B \int \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{123}$$

$$\times \prod_{x \in A} \delta(\phi(x,0) - \phi'_A(x)) \delta(\phi(x,\beta) - \phi_A(x)) \tag{124}$$

$$\times \prod_{x \in B} \delta(\phi(x,0) - \phi_B(x)) \delta(\phi(x,\beta) - \phi_B(x)) \tag{125}$$

**Key identity:**

$$\int d\phi_x \, \delta(\phi(x,0) - \phi_x) \delta(\phi(x,\beta) - \phi_x) = \delta(\phi(x,0) - \phi(x,\beta)) \tag{126}$$

**Result:**

$$\langle \phi_A | \rho_A | \phi'_A \rangle = \frac{1}{Z} \int \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{127}$$

$$\times \prod_{x \in A} \delta(\phi(x,0) - \phi'_A(x)) \delta(\phi(x,\beta) - \phi_A(x)) \tag{128}$$

$$\times \prod_{x \in B} \delta(\phi(x,0) - \phi(x,\beta)) \tag{129}$$

**Geometric interpretation:**

- Region $A$: Open cut (matrix indices $\phi_A$ and $\phi'_A$)
- Region $B$: Sewn together (periodic boundary condition)

**Boundary conditions summary:**
Geometry: Cylinder with cuts along region $A$.

| Region | Boundary Condition | Geometry |
|---|---|---|
| $A$ | $\phi(x,0) = \phi'_A$, $\phi(x,\beta) = \phi_A$ | Open cut (matrix indices) |
| $B$ | $\phi(x,0) = \phi(x,\beta)$ | Sewn together (periodic) |

## 3.9 Comparison: Discrete vs. Continuum Partial Trace

To clarify the path integral derivation, we compare the standard discrete partial trace from quantum mechanics with the functional integral over region $B$.

### 3.9.1 Standard Discrete Partial Trace

Consider a bipartite system with Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, where $\mathcal{H}_A$ and $\mathcal{H}_B$ are finite-dimensional. Let $\{|a\rangle\}$ be an orthonormal basis for $\mathcal{H}_A$ and $\{|b\rangle\}$ for $\mathcal{H}_B$. The full basis is $\{|a,b\rangle = |a\rangle \otimes |b\rangle\}$.

A general density matrix $\rho$ can be expanded as:

$$\rho = \sum_{a,b,a',b'} \rho_{ab,a'b'} \, |a,b\rangle\langle a',b'| \tag{130}$$

The reduced density matrix for subsystem $A$ is obtained by tracing over $B$:

$$\rho_A = \mathrm{Tr}_B(\rho) = \sum_b \langle b|\rho|b\rangle_B \tag{131}$$

where the subscript on the inner product indicates the partial inner product over $B$ only. In terms of matrix elements:

$$(\rho_A)_{aa'} = \langle a|\rho_A|a'\rangle = \sum_b \langle a,b|\rho|a',b\rangle = \sum_b \rho_{ab,a'b'} \tag{D1}$$

**Key observation:** The partial trace sums over the diagonal elements of subsystem $B$ in any chosen basis. The indices corresponding to $B$ are set equal in the bra and ket before summation.

### 3.9.2 Path Integral Partial Trace (Continuum Version)

In quantum field theory, the basis states are field configurations $\phi_A$ on $A$ and $\phi_B$ on $B$. The partial trace is:

$$\langle \phi_A|\rho_A|\phi'_A\rangle = \int \mathcal{D}\phi_B \, \langle \phi_A, \phi_B|\rho|\phi'_A, \phi_B\rangle \tag{C1}$$

This is the exact continuous analog of (D1):

### 3.9.3 From Trace to Sewing: The Key Delta Function Identity

In the discrete case, we can write the sum over $b$ using a Kronecker delta to emphasize the equality of indices:

$$\sum_b \delta_{b,b_1}\delta_{b,b_2} = \delta_{b_1,b_2} \tag{D2}$$

| **Discrete (D1)** | **Continuous (C1)** |
|---|---|
| Sum over basis index $b$ | Functional integral over field $\phi_B$ |
| Matrix element $\rho_{ab,a'b}$ | Kernel $\langle \phi_A, \phi_B \vert \rho \vert \phi'_A, \phi_B \rangle$ |
| Summation $\sum_b$ | Integration $\int \mathcal{D}\phi_B$ |

This trivial identity is the discrete analog of the functional identity used in the path integral derivation:

$$\int \mathcal{D}\phi_B \prod_{\mathbf{x}\in B} \delta\big(\phi(\mathbf{x},0) - \phi_B(\mathbf{x})\big)\delta\big(\phi(\mathbf{x},\beta) - \phi_B(\mathbf{x})\big) = \prod_{\mathbf{x}\in B} \delta\big(\phi(\mathbf{x},0) - \phi(\mathbf{x},\beta)\big) \tag{C2}$$

Both equations state: *summing over a shared index forces the two appearances of that index to be equal.*

### 3.9.4 Geometric Interpretation Comparison

**Discrete case:**

- We start with a density matrix that has independent bra and ket indices for $B$: $\rho_{ab,a'b'}$
- The partial trace sets $b = b'$ and sums over $b$
- Visual: The two "legs" of the matrix for subsystem $B$ are connected (traced together)

**Continuum path integral case:**

- The thermal density matrix is represented by a path integral on a cylinder of circumference $\beta$, with independent boundary conditions at $\tau = 0$ (bra) and $\tau = \beta$ (ket)
- On region $A$, the bra and ket boundary values $\phi'_A$ and $\phi_A$ remain independent (matrix indices)
- On region $B$, the partial trace forces $\phi_B$ at $\tau = 0$ and $\tau = \beta$ to be equal, and then integrates over $\phi_B$
- This is exactly sewing the two ends of the cylinder together on $B$, making the time direction periodic there

### 3.9.5 Lattice Version as Direct Bridge

The discrete lattice formulation makes the correspondence explicit. Let spatial lattice sites be partitioned into $A$ and $B$. The density matrix is $\rho(\{\phi_i\}, \{\phi'_i\})$.

The discrete partial trace is:

$$\rho_A(\{\phi_A\}, \{\phi'_A\}) = \int \prod_{j\in B} d\phi_j \; \rho\big(\{\phi_A, \phi_B\}, \{\phi'_A, \phi_B\}\big) \tag{L1}$$

When $\rho$ is expressed via a lattice path integral (with time slices $\tau = 0, \ldots, N_\tau$), we get:

- On sites in $B$: $\sum_{\phi_j} \delta_{\phi(j,0),\phi_j} \delta_{\phi(j,N_\tau),\phi_j} = \delta_{\phi(j,0),\phi(j,N_\tau)}$
- On sites in $A$: $\delta_{\phi(i,0),\phi'_A(i)} \delta_{\phi(i,N_\tau),\phi_A(i)}$

This is the perfect lattice parallel to the continuous delta functional identity.

### 3.9.6 Why This Matters for Entanglement Entropy

The reduced density matrix $\rho_A$ is the central object for computing entanglement entropy $S_A = -\operatorname{Tr}(\rho_A \log \rho_A)$. In the replica trick, one computes $\operatorname{Tr}(\rho_A^n)$ by taking $n$ copies of the cut cylinder and cyclically gluing them along the cuts on $A$.

- The sewing on $B$ comes from the partial trace over $B$
- The cyclic gluing on $A$ comes from the matrix multiplication in $\rho_A^n$

Thus, understanding the partial trace as a sewing operation on $B$ is essential for the geometric computation of entanglement entropy in QFT.

### 3.9.7 Summary Table

| **Aspect** | **Discrete QM Partial Trace** | **QFT Path Integral Partial Trace** |
|---|---|---|
| Trace operation | $\sum_b \langle b\|\rho\|b\rangle$ | $\int \mathcal{D}\phi_B \langle \phi_A, \phi_B \|\rho\| \phi'_A, \phi_B\rangle$ |
| Index setting | $b = b'$ (bra and ket index equal) | $\phi_B$ same at $\tau = 0$ and $\tau = \beta$ |
| Delta identity | $\sum_b \delta_{b,b_1}\delta_{b,b_2} = \delta_{b_1,b_2}$ | $\int \mathcal{D}\phi_B \delta(\phi(0) - \phi_B)\delta(\phi(\beta) - \phi_B) = \delta(\phi(0) - \phi(\beta))$ |
| Geometric result | Connect subsystem $B$ legs | Sew cylinder on region $B$ (periodic boundary) |

The path integral partial trace is the natural functional generalization of the standard discrete partial trace, with the sum over a complete set of states replaced by a functional integral and the index equality enforced by a Dirac delta functional.

## 3.10 Density Matrix as Path Integral

The thermal density matrix at temperature $\beta^{-1}$ is:

$$\rho = \frac{e^{-\beta H}}{Z}, \quad Z = \operatorname{Tr}\left(e^{-\beta H}\right) \tag{132}$$

In path integral form:

$$\rho(\{\phi_x\}, \{\phi'_x\}) = \frac{1}{Z} \int \mathcal{D}\phi \, e^{-S_E[\phi]} \prod_x \delta(\phi(x,0) - \phi'_x)\delta(\phi(x,\beta) - \phi_x) \tag{133}$$

where $S_E$ is the Euclidean action.

## 3.11 Reduced Density Matrix: Tracing Out Region $B$

To obtain $\rho_A = \mathrm{Tr}_B\, \rho$, we integrate over field configurations in region $B$:

$$\rho_A(\{\phi_A\}, \{\phi'_A\}) = \int \mathcal{D}\phi_B\, \rho(\{\phi_A, \phi_B\}, \{\phi'_A, \phi_B\}) \tag{134}$$

**Geometric interpretation:** The path integral is on a cylinder with:

- Region $B$: sewn together (periodic boundary condition)
- Region $A$: cut open (boundary conditions $\phi_A$ and $\phi'_A$)

## 3.12 Replica Construction

The trace $\mathrm{Tr}(\rho_A^n)$ can be expressed as a product of matrix elements.

**Step 1:** Start with trace definition. From Eq. (15):

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi\, \langle\phi|\rho_A^n|\phi\rangle \tag{135}$$

**Step 2:** Expand $\rho_A^n$ as operator product:

$$\rho_A^n = \underbrace{\rho_A \cdot \rho_A \cdot \cdots \cdot \rho_A}_{n \text{ factors}} \tag{136}$$

Therefore:

$$\langle\phi|\rho_A^n|\phi\rangle = \langle\phi|\rho_A \cdot \rho_A \cdot \cdots \cdot \rho_A|\phi\rangle \tag{137}$$

**Step 3:** Insert identity between each factor. Completeness relation:

$$I = \int \mathcal{D}\phi'\, |\phi'\rangle\langle\phi'| \tag{138}$$

For $n = 2$:

$$\langle\phi|\rho_A^2|\phi\rangle = \langle\phi|\rho_A \cdot \rho_A|\phi\rangle \tag{139}$$

$$= \langle\phi|\rho_A \left(\int \mathcal{D}\phi^{(2)}\, |\phi^{(2)}\rangle\langle\phi^{(2)}|\right) \rho_A|\phi\rangle \tag{140}$$

$$= \int \mathcal{D}\phi^{(2)}\, \langle\phi|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi\rangle \tag{141}$$

For $n = 3$:

$$\langle\phi|\rho_A^3|\phi\rangle = \langle\phi|\rho_A \cdot \rho_A \cdot \rho_A|\phi\rangle \tag{142}$$

$$= \int \mathcal{D}\phi^{(2)}\mathcal{D}\phi^{(3)}\, \langle\phi|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle\langle\phi^{(3)}|\rho_A|\phi\rangle \tag{143}$$

For general $n$:

$$\langle\phi|\rho_A^n|\phi\rangle = \int \mathcal{D}\phi^{(2)} \cdots \mathcal{D}\phi^{(n)}\, \langle\phi|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle \cdots \langle\phi^{(n)}|\rho_A|\phi\rangle \tag{144}$$

**Step 4:** Relabel for cyclic symmetry. Define $\phi^{(1)} \equiv \phi$:

$$\langle\phi^{(1)}|\rho_A^n|\phi^{(1)}\rangle = \int \mathcal{D}\phi^{(2)} \cdots \mathcal{D}\phi^{(n)}\, \langle\phi^{(1)}|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle \cdots \langle\phi^{(n)}|\rho_A|\phi^{(1)}\rangle \tag{145}$$

### 3.12.1 Clarification: Replica Indices vs. Time Slices

**Important:** The superscripts $(1), (2), \ldots, (n)$ label *replicas* (copies of the system), NOT time slices.

| **Replica Indices** | **Time Slices** |
|---|---|
| $\phi^{(1)}, \phi^{(2)}, \ldots, \phi^{(n)}$ | $\tau_1, \tau_2, \ldots, \tau_{N_\tau}$ |
| $n$ independent copies of the *entire* system | One system discretized in Euclidean time |
| Used for computing $\mathrm{Tr}(\rho_A^n)$ (replica trick) | Used for defining the path integral |
| Each replica has its own time direction $\tau \in [0, \beta]$ | Time slices live within one replica |
| Cyclic gluing: copy $j$ connects to copy $j+1$ along region $A$ | Time evolution: $\tau_k$ connects to $\tau_{k+1}$ |

**Visual distinction:**

- **Replicas:** Imagine $n$ separate cylinders (each representing a full path integral from $\tau = 0$ to $\tau = \beta$). We glue them together along the cut in region $A$ to form an $n$-sheeted Riemann surface.
- **Time slices:** Within *each* cylinder, we discretize the time direction into $N_\tau$ slices to define the functional measure $\mathcal{D}\phi = \prod_{k=1}^{N_\tau} d\phi_k$.

**Why the confusion arises:** Both involve "copies" in some sense:

- Time slices: We copy the field variable at each time step $\phi(\tau_k)$
- Replicas: We copy the *entire system* (all time slices together) $n$ times

**Key insight:** The replica index $(j)$ labels which *sheet* of the Riemann surface we are on. When we integrate over $\phi^{(j)}$, we are gluing sheet $j-1$ to sheet $j$ along region $A$. This is completely independent of the time discretization within each sheet.

**Example for $n = 3$:**

$$\mathrm{Tr}(\rho_A^3) = \int \mathcal{D}\phi^{(1)}\mathcal{D}\phi^{(2)}\mathcal{D}\phi^{(3)} \, \langle\phi^{(1)}|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle\langle\phi^{(3)}|\rho_A|\phi^{(1)}\rangle \tag{146}$$

$$= \int \mathcal{D}\phi^{(1)}\mathcal{D}\phi^{(2)}\mathcal{D}\phi^{(3)} \underbrace{\left(\int_{\phi^{(2)}}^{\phi^{(1)}} \mathcal{D}\phi \, e^{-S_E}\right)}_{\text{Replica 1}} \underbrace{\left(\int_{\phi^{(3)}}^{\phi^{(2)}} \mathcal{D}\phi \, e^{-S_E}\right)}_{\text{Replica 2}} \underbrace{\left(\int_{\phi^{(1)}}^{\phi^{(3)}} \mathcal{D}\phi \, e^{-S_E}\right)}_{\text{Replica 3}} \tag{147}$$

Each replica integral contains its own time slices $\tau_1, \ldots, \tau_{N_\tau}$, but the replica index $(j)$ tells us which copy of the cylinder we are on.

**Step 5:** Substitute back into trace:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi^{(1)} \, \langle\phi^{(1)}|\rho_A^n|\phi^{(1)}\rangle \tag{148}$$

$$= \int \mathcal{D}\phi^{(1)}\mathcal{D}\phi^{(2)}\cdots\mathcal{D}\phi^{(n)} \, \langle\phi^{(1)}|\rho_A|\phi^{(2)}\rangle\langle\phi^{(2)}|\rho_A|\phi^{(3)}\rangle\cdots\langle\phi^{(n)}|\rho_A|\phi^{(1)}\rangle \tag{149}$$

**Final result:**

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi^{(1)} \cdots \mathcal{D}\phi^{(n)} \, \langle \phi^{(1)} | \rho_A | \phi^{(2)} \rangle \langle \phi^{(2)} | \rho_A | \phi^{(3)} \rangle \cdots \langle \phi^{(n)} | \rho_A | \phi^{(1)} \rangle \tag{150}$$

**Geometric picture:**

- Each matrix element $\langle \phi^{(j)} | \rho_A | \phi^{(j+1)} \rangle$ represents one copy of the system
- The cyclic structure connects copy $j$ to copy $j+1$
- Integration over all $\phi^{(k)}$ sews the copies together
- Result: $n$-sheeted Riemann surface $\mathcal{R}_n$

## 3.13 Partition Function Ratio

The key result is:

$$\mathrm{Tr}(\rho_A^n) = \frac{Z_n}{(Z_1)^n} \tag{151}$$

where:

- $Z_n$: partition function on the $n$-sheeted surface $\mathcal{R}_n$
- $Z_1$: partition function on a single copy (normalized to 1)

### 3.13.1 Derivation of the Ratio

**Step 1: Definition of partition functions**
The partition function on a manifold $\mathcal{M}$ is:

$$Z[\mathcal{M}] = \int_{\mathcal{M}} \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{152}$$

For the replica trick:

- $Z_1 = Z[\mathbb{C}]$: partition function on the complex plane (single sheet)
- $Z_n = Z[\mathcal{R}_n]$: partition function on the $n$-sheeted Riemann surface

**Step 2: Normalization of the density matrix**
The thermal density matrix is normalized:

$$\mathrm{Tr}(\rho) = 1 \quad \Rightarrow \quad \frac{1}{Z_1} \int \mathcal{D}\phi \, e^{-S_E[\phi]} = 1 \tag{153}$$

Therefore, the normalized density matrix element is:

$$\rho(\phi_f | \phi_i) = \frac{1}{Z_1} \int_{\phi_i}^{\phi_f} \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{154}$$

**Step 3: Product of density matrices**
From the replica construction (Eq. 59):

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi^{(1)} \cdots \mathcal{D}\phi^{(n)}\, \rho_A(\phi^{(1)}|\phi^{(2)})\rho_A(\phi^{(2)}|\phi^{(3)}) \cdots \rho_A(\phi^{(n)}|\phi^{(1)}) \tag{155}$$

Substitute the path integral form:

$$\rho_A(\phi^{(j)}|\phi^{(j+1)}) = \frac{1}{Z_1} \int_{\phi^{(j+1)}}^{\phi^{(j)}} \mathcal{D}\phi^{(j)}\, e^{-S_E[\phi^{(j)}]} \tag{156}$$

where the path integral is over fields on copy $j$ with boundary conditions matching region $A$.

**Step 4: Combine the path integrals**
Multiplying $n$ factors:

$$\mathrm{Tr}(\rho_A^n) = \int \mathcal{D}\phi^{(1)} \cdots \mathcal{D}\phi^{(n)} \prod_{j=1}^{n} \left[ \frac{1}{Z_1} \int_{\phi^{(j+1)}}^{\phi^{(j)}} \mathcal{D}\phi^{(j)}\, e^{-S_E[\phi^{(j)}]} \right] \tag{157}$$

$$= \frac{1}{(Z_1)^n} \int \mathcal{D}\phi^{(1)} \cdots \mathcal{D}\phi^{(n)}\, e^{-\sum_{j=1}^{n} S_E[\phi^{(j)}]} \tag{158}$$

**Step 5: Cyclic gluing**
The cyclic boundary condition $\phi^{(n+1)} \equiv \phi^{(1)}$ means:

- Copy 1 connects to copy 2 along region $A$
- Copy 2 connects to copy 3 along region $A$
- $\cdots$
- Copy $n$ connects back to copy 1 along region $A$

This creates the $n$-sheeted Riemann surface $\mathcal{R}_n$.
**Step 6: Identify $Z_n$**
The combined path integral is precisely the partition function on $\mathcal{R}_n$:

$$\int_{\mathcal{R}_n} \mathcal{D}\phi\, e^{-S_E[\phi]} = Z_n \tag{159}$$

**Final result:**

$$\boxed{\mathrm{Tr}(\rho_A^n) = \frac{Z_n}{(Z_1)^n}} \tag{160}$$

**Physical interpretation:**

- Numerator $Z_n$: path integral on $n$-sheeted surface (replica manifold)
- Denominator $(Z_1)^n$: normalization from $n$ copies of the density matrix
- The ratio is finite and well-defined even though individual path integrals diverge

# 4 Twist Fields

## 4.1 Uniformizing Map

The $n$-sheeted Riemann surface $\mathcal{R}_n$ can be mapped to the complex plane via a conformal transformation.

**Single interval $A = [u, v]$:** The uniformizing map is:

$$w = \left(\frac{z-u}{z-v}\right)^{1/n} \tag{161}$$

where:

- $z$: coordinate on the $n$-sheeted surface $\mathcal{R}_n$
- $w$: coordinate on the complex plane $\mathbb{C}$

**Properties:**

- Branch points at $z = u$ and $z = v$ map to $w = 0$ and $w = \infty$
- The $n$ sheets are unwrapped into the full $w$-plane
- The map is $n$-to-1 except at branch points

## 4.2 Stress Tensor Transformation

Under a conformal transformation $z \to w(z)$, the stress tensor transforms as:

$$T(z) = \left(\frac{dw}{dz}\right)^2 T(w) + \frac{c}{12}\{w, z\} \tag{162}$$

where $\{w, z\}$ is the Schwarzian derivative:

$$\{w, z\} = \frac{w'''}{w'} - \frac{3}{2}\left(\frac{w''}{w'}\right)^2 \tag{163}$$

**Compute derivatives of $w(z)$:**

$$w' = \frac{dw}{dz} = \frac{1}{n}\left(\frac{z-u}{z-v}\right)^{\frac{1}{n}-1} \frac{v-u}{(z-v)^2} \tag{164}$$

$$w'' = \frac{d^2w}{dz^2} = \frac{1}{n}\left(\frac{1}{n}-1\right)\left(\frac{z-u}{z-v}\right)^{\frac{1}{n}-2} \frac{(v-u)^2}{(z-v)^4} - \frac{2}{n}\left(\frac{z-u}{z-v}\right)^{\frac{1}{n}-1} \frac{v-u}{(z-v)^3} \tag{165}$$

**Schwarzian derivative (after simplification):**

$$\{w, z\} = \frac{n^2-1}{2n^2} \frac{(v-u)^2}{(z-u)^2(z-v)^2} \tag{166}$$

## 4.3 Scaling Dimension of Twist Fields

The expectation value of the stress tensor in the presence of twist fields is:

$$\langle T(z)\rangle_{\mathcal{R}_n} = \frac{c}{12}\{w, z\} \tag{167}$$

**Compare to CFT Ward identity:** For primary fields $\mathcal{T}_n(u)$ and $\tilde{\mathcal{T}}_n(v)$:

$$\langle T(z)\mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = \left[\frac{h_n}{(z-u)^2} + \frac{h_n}{(z-v)^2} + \frac{1}{z-u}\partial_u + \frac{1}{z-v}\partial_v\right]\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle \tag{168}$$

**Match the singular terms:** Comparing Eq. (47) with the Schwarzian (Eq. 46):

$$\frac{c}{12}\cdot\frac{n^2-1}{2n^2}\frac{(v-u)^2}{(z-u)^2(z-v)^2} = \frac{h_n}{(z-u)^2} + \frac{h_n}{(z-v)^2} + \cdots \tag{169}$$

**Extract scaling dimension:**

$$h_n = \frac{c}{24}\left(n - \frac{1}{n}\right) \tag{170}$$

The full scaling dimension (holomorphic + antiholomorphic) is:

$$\Delta_n = h_n + \bar{h}_n = \frac{c}{12}\left(n - \frac{1}{n}\right) \tag{171}$$

## 4.4 Two-Point Function of Twist Fields

For primary fields with scaling dimension $\Delta_n$ on the plane:

$$\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = \frac{c_n}{|u-v|^{2\Delta_n}} \tag{172}$$

where $c_n$ is a non-universal constant.

**Substitute $\Delta_n$:**

$$\langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = c_n|u-v|^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} \tag{173}$$

**Relate to partition function:**

$$\frac{Z_n}{(Z_1)^n} = \langle \mathcal{T}_n(u)\tilde{\mathcal{T}}_n(v)\rangle = c_n\left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} \tag{174}$$

where $L = |u-v|$ and $a$ is the UV cutoff (lattice spacing).

# 5 The $n \to 1$ Limit

## 5.1 Rényi Entropy

From Eq. (52):

$$\mathrm{Tr}(\rho_A^n) = c_n \left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} \tag{175}$$

**Rényi entropy:**

$$S_A^{(n)} = \frac{1}{1-n} \ln \mathrm{Tr}(\rho_A^n) \tag{176}$$

$$= \frac{1}{1-n}\left[\ln c_n - \frac{c}{6}\left(n - \frac{1}{n}\right)\ln\left(\frac{L}{a}\right)\right] \tag{177}$$

$$= \frac{c}{6}\left(1+\frac{1}{n}\right)\ln\left(\frac{L}{a}\right) + \frac{\ln c_n}{1-n} \tag{178}$$

Define $c_n' = \frac{\ln c_n}{1-n}$ (non-universal constant).

## 5.2 Von Neumann Entropy

**Take the limit $n \to 1$:**

$$S_A = \lim_{n\to 1} S_A^{(n)} = \frac{c}{3}\ln\left(\frac{L}{a}\right) + c_1' \tag{179}$$

**Alternative derivation via L'Hôpital's rule:**
From the replica trick formula:

$$S_A = -\left.\frac{\partial}{\partial n}\mathrm{Tr}(\rho_A^n)\right|_{n=1} \tag{180}$$

Compute the derivative:

$$\frac{\partial}{\partial n}\mathrm{Tr}(\rho_A^n) = \frac{\partial}{\partial n}\left[c_n\left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)}\right] \tag{181}$$

$$= \left(\frac{\partial c_n}{\partial n}\right)\left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} + c_n\left(\frac{L}{a}\right)^{-\frac{c}{6}\left(n-\frac{1}{n}\right)} \cdot \left(-\frac{c}{6}\right)\left(1+\frac{1}{n^2}\right)\ln\left(\frac{L}{a}\right) \tag{182}$$

Evaluate at $n = 1$:

$$\left.\frac{\partial}{\partial n}\mathrm{Tr}(\rho_A^n)\right|_{n=1} = \left.\frac{\partial c_n}{\partial n}\right|_{n=1} - \frac{c}{3}\ln\left(\frac{L}{a}\right) \tag{183}$$

**Therefore:**

$$S_A = -\left.\frac{\partial c_n}{\partial n}\right|_{n=1} + \frac{c}{3}\ln\left(\frac{L}{a}\right) = \frac{c}{3}\ln\left(\frac{L}{a}\right) + \mathrm{const} \tag{184}$$

## 5.3 Main Result

**Entanglement Entropy in 1+1D CFT**
For a subsystem of length $L$ in the vacuum state of a 1+1D CFT:

$$S_A = \frac{c}{3} \ln \left( \frac{L}{a} \right) + \text{const} \tag{185}$$

where:

- $c$: central charge of the CFT
- $L$: length of subsystem $A$
- $a$: UV cutoff (lattice spacing)

# 6 Numerical Method

## 6.1 Correlation Matrix Approach

For free fermion systems (e.g., transverse Ising model), the entanglement entropy can be computed from the correlation matrix.

**Correlation matrix:**

$$C_{ij} = \langle c_i^\dagger c_j \rangle \quad \text{for } i, j \in A \tag{186}$$

**Entanglement entropy:**

$$S_A = -\operatorname{Tr}\left[C \ln C + (I - C)\ln(I - C)\right] \tag{187}$$

where the trace is over the subsystem $A$.

**Scaling with subsystem size:**

$$S_A(L) = \frac{c}{3}\ln\left(\frac{L}{a}\right) + \text{const} \tag{188}$$

**Extracting central charge:** Fit numerical data to:

$$S_A(L) = \frac{c}{3}\log_2\left(\frac{L}{a}\right) + \text{const} \tag{189}$$

The slope gives $c/3$, so $c = 3 \times \text{slope}$.

**For transverse Ising model at critical point ($g = J$):**

- Expected: $c = 1/2$
- Expected slope: $c/3 = 1/6 \approx 0.1667$

# Summary

## 6.2 Key Equations

| Equation | Formula | Meaning |
|---|---|---|
| Path integral | $\langle\phi_f\|e^{-\beta H}\|\phi_i\rangle = \int \mathcal{D}\phi\, e^{-S_E}$ | Transition amplitude |
| Density matrix | $\rho(\phi_f\|\phi_i) = \frac{1}{Z}\int \mathcal{D}\phi\, e^{-S_E}$ | Thermal state |
| Reduced $\rho_A$ | Cylinder with cuts along $A$ | Partial trace |
| Replica trick | $\mathrm{Tr}(\rho_A^n) = Z_n(A)/Z^n$ | $n$-sheeted surface |
| Twist dimension | $\Delta_n = \frac{c}{12}(n - 1/n)$ | From stress tensor |
| Two-point function | $\langle\mathcal{T}_n\tilde{\mathcal{T}}_n\rangle = L^{-2\Delta_n}$ | CFT on plane |
| Rényi entropy | $S_A^{(n)} = \frac{c}{6}(1 + 1/n)\ln(L/a)$ | Before $n \to 1$ limit |
| Von Neumann | $S_A = \frac{c}{3}\ln(L/a)$ | Final result |

## 6.3 Derivation Steps

| Step | Operation | Result |
|---|---|---|
| 1 | Trotter decomposition | $e^{-\beta H} = (e^{-\epsilon H})^N$ |
| 2 | Complete sets of states | $\int dq_k\|q_k\rangle\langle q_k\|$ |
| 3 | Momentum eigenstates | $\langle q\|p\rangle = e^{ipq}$ |
| 4 | Gaussian integrals | Kinetic term $\frac{1}{2}m\dot{q}^2$ |
| 5 | Continuum limit | Path integral $\int \mathcal{D}\phi\, e^{-S_E}$ |
| 6 | Delta functions | Boundary conditions |
| 7 | Partial trace | Cylinder with cuts |
| 8 | Replica trick | $n$-sheeted surface |
| 9 | Twist fields | CFT correlators |
| 10 | $n \to 1$ limit | $S_A = \frac{c}{3}\ln(L/a)$ |

## 6.4 Generalizations

| Situation | Formula |
|---|---|
| Infinite system, $T = 0$ | $S = \frac{c}{3}\ln(L/a)$ |
| Finite temperature | $S = \frac{c}{3}\ln\left[\frac{\beta}{\pi a}\sinh\left(\frac{\pi L}{\beta}\right)\right]$ |
| Finite size (periodic) | $S = \frac{c}{3}\ln\left[\frac{L}{\pi a}\sin\left(\frac{\pi\ell}{L}\right)\right]$ |
| After global quench | $S(t) \sim \frac{\pi c}{3\beta}t$ (linear growth) |
| KZM quench | $S_\infty = \frac{c}{6}\ln(\tau_Q \ln\tau_Q)$ |

# Acknowledgments

This document follows the Calabrese-Cardy derivation of entanglement entropy in CFT. For further reading:


- Calabrese & Cardy (2004). "Entanglement entropy and quantum field theory" J. Stat. Mech. P06002
- Calabrese & Cardy (2009). "Entanglement entropy and conformal field theory" J. Phys. A 42 504005
- Holzhey, Larsen, Wilczek (1994). "Geometric and renormalized entropy in conformal field theory" Nucl. Phys. B424 443

# Appendix A: Position-Momentum Overlap

**Goal:** Derive $\langle p|q\rangle = e^{-ipq}$ from first principles.

## Conventions

We work in units where $\hbar = 1$. The position and momentum operators satisfy:

$$[\hat{q}, \hat{p}] = i \tag{190}$$

Position eigenstates:

$$\hat{q}|q\rangle = q|q\rangle, \quad \langle q|q'\rangle = \delta(q - q') \tag{191}$$

Momentum eigenstates:

$$\hat{p}|p\rangle = p|p\rangle, \quad \langle p|p'\rangle = 2\pi\delta(p - p') \tag{192}$$

The $2\pi$ factor is a convention choice (alternative: $\langle p|p'\rangle = \delta(p - p')$ with different Fourier conventions).

## Derivation

**Step 1: Translation operator**

The momentum operator generates translations in position:

$$e^{-ia\hat{p}}|q\rangle = |q + a\rangle \tag{193}$$

*Proof:* Apply $\hat{q}$ to both sides:

$$\begin{aligned}
\hat{q}e^{-ia\hat{p}}|q\rangle &= e^{-ia\hat{p}}(e^{ia\hat{p}}\hat{q}e^{-ia\hat{p}})|q\rangle && (194)\\
&= e^{-ia\hat{p}}(\hat{q} + a)|q\rangle \quad \text{(using } [\hat{q}, \hat{p}] = i) && (195)\\
&= e^{-ia\hat{p}}(q + a)|q\rangle && (196)\\
&= (q + a)e^{-ia\hat{p}}|q\rangle && (197)
\end{aligned}$$

Thus $e^{-ia\hat{p}}|q\rangle$ is an eigenstate of $\hat{q}$ with eigenvalue $q + a$, so it equals $|q + a\rangle$ (up to phase, which we set to 1).

**Step 2: Matrix element of translation**

Consider $\langle p|q\rangle$. Apply the translation operator:

$$\begin{aligned}
\langle p|q + a\rangle &= \langle p|e^{-ia\hat{p}}|q\rangle && (198)\\
&= e^{-iap}\langle p|q\rangle \quad \text{(since } \hat{p}|p\rangle = p|p\rangle) && (199)
\end{aligned}$$

**Step 3: Differential equation**

Expand for infinitesimal $a$:

$$\langle p|q + a\rangle = \langle p|q\rangle + a\frac{\partial}{\partial q}\langle p|q\rangle + O(a^2) \tag{200}$$

$$e^{-iap}\langle p|q\rangle = (1 - iap + O(a^2))\langle p|q\rangle \tag{201}$$

Matching $O(a)$ terms:

$$\frac{\partial}{\partial q}\langle p|q\rangle = -ip\langle p|q\rangle \tag{202}$$

**Step 4: Solve the differential equation**

The solution is:

$$\langle p|q\rangle = C(p)e^{-ipq} \tag{203}$$

where $C(p)$ is a $q$-independent normalization constant.

**Step 5: Determine normalization**

Use the completeness relation for position states:

$$\langle p|p'\rangle = \int dq\, \langle p|q\rangle\langle q|p'\rangle \tag{204}$$

Substitute $\langle p|q\rangle = C(p)e^{-ipq}$ and $\langle q|p'\rangle = \langle p'|q\rangle^* = C^*(p')e^{ip'q}$:

$$\langle p|p'\rangle = \int dq\, C(p)e^{-ipq}C^*(p')e^{ip'q} \tag{205}$$

$$= C(p)C^*(p')\int dq\, e^{i(p'-p)q} \tag{206}$$

$$= C(p)C^*(p')\cdot 2\pi\delta(p'-p) \tag{207}$$

We require $\langle p|p'\rangle = 2\pi\delta(p-p')$, so:

$$C(p)C^*(p') = 1 \quad \Rightarrow \quad |C(p)| = 1 \tag{208}$$

We choose the phase convention $C(p) = 1$, giving:

$$\boxed{\langle p|q\rangle = e^{-ipq}} \tag{209}$$

## Alternative Convention

Some texts use symmetric normalization:

$$\langle p|q\rangle = \frac{1}{\sqrt{2\pi}}e^{-ipq} \tag{210}$$

with $\langle p|p'\rangle = \delta(p-p')$ and completeness $\int dp\, |p\rangle\langle p| = I$.

Our convention ($\langle p|q\rangle = e^{-ipq}$) is common in path integral literature and simplifies formulas by absorbing $2\pi$ factors into the momentum measure $\int \frac{dp}{2\pi}$.